Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2021-P1-Model)** **Date: 10-05-2026**

Time: 09.00Am to 12.00Pm **GTA-31** **Max. Marks: 180**

KEY SHEET

PHYSICS

1	**B**	2	**C**	3	**A**	4	**B**	5	**1.33**	6	**188.4 - 189.1**
7	**6**	8	**6**	9	**1**	10	**3**	11	**AC**	12	**ABC**
13	**BCD**	14	**D**	15	**BC**	16	**AD**	17	**6**	18	**1**
19	**2**										

CHEMISTRY

20	**D**	21	**B**	22	**A**	23	**D**	24	**9**	25	**10**
26	**14**	27	**5**	28	**5**	29	**7**	30	**C**	31	**A**
32	**ACD**	33	**ABC**	34	**ABD**	35	**ABCD**	36	**18**	37	**3**
38	**4**										

MATHEMATICS

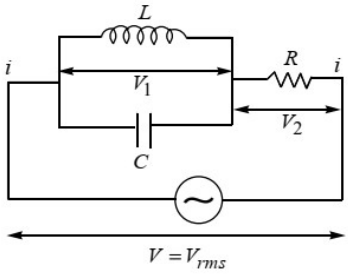
39	**A**	40	**A**	41	**D**	42	**D**	43	**2**	44	**6**
45	**33**	46	**0.28 - 0.29**	47	**3**	48	**1**	49	**ABCD**	50	**AD**
51	**ABC**	52	**AD**	53	**ABC**	54	**BC**	55	**80**	56	**9**
57	**4**										

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SOLUTIONS

PHYSICS

1. $$V_1^2+V_2^2=V^2 \quad \frac{V_1}{X_L}-\frac{V_1}{X_C}=iV_1\left(\frac{1}{\omega L}-\omega C\right)=i$$



$$V_1=\frac{\omega Li}{1-\omega^2LC}V_2=iR$$

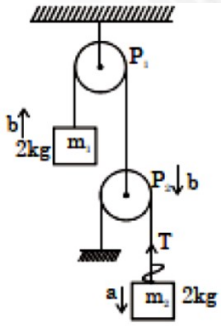
$$\frac{\omega^2L^2}{\left(1-\omega^2LC\right)^2}+R^2=\frac{V^2}{i^2}i=\frac{V}{\sqrt{R^2+\frac{\omega^2L^2}{\left(1-\omega^2LC\right)^2}}}$$

2. $$2gR\left(1-\cos\theta\right)+u^2=g\cos\theta R$$

$$u^2+\frac{gR}{3}=\frac{5}{6}gR \qquad u^2=\frac{gR}{2} \qquad V^2=\frac{5gR}{2}$$

3. We have

$$2T-20=2b\#(i) \quad 20-T=2a\#(ii)$$



$$a-2b=0 \qquad 2T-20=2b$$

$$40-2T=8b\Rightarrow a=2b\ldots..(iii)$$

$$20=10b\Rightarrow b=2m/s^2$$

$$Also\,T=12N \qquad \therefore V_{wave}=\sqrt{\frac{12}{3\times10^{-2}}}=20m/s$$

$$a_{pulley}=2m/s^2 \quad \therefore 0.2=20t-\frac{1}{2}2t^2\Rightarrow t^2-20t+0.2=0t=0.01s$$

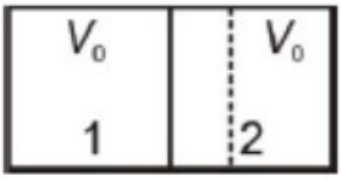
4. When all particles on wire at mean position, the induced emf will be maximum because velocity is maximum at mean position. $=B(a\omega)dx$

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Where $a = A\sin\dfrac{\pi x}{L}$

5, 6.

Let the cross sectional area be a.



Then the work done from a displacement of x to $xadx$ is $(P_2 - P_1)dV$

$P_1(V_0 + ax) = P_2(V_0 - ax) = P_0V_0$

$P_1 = \dfrac{P_0V_0}{V_0 + ax}, P_2 = \dfrac{P_0V_0}{V_0 - ax}$

$P_2 - P_1 = P_0V_0\left(\dfrac{V_0 + ax - V_0 + ax}{V_0^2 - a^2x^2}\right) = \dfrac{2axP_0V_0}{V_0^2 - a^2x^2}$

$W = \int(P_2 - P_1)dV = \int(P_2 - P_1)adx \qquad = \int\dfrac{2a^2P_0V_0xdx}{V_0^2 - a^2x^2}$

$V_0^2 - a^2x^2 = b \Rightarrow -a^2 2xdx = db$

$\dfrac{a^2P_0V_0}{-a^2}\int\dfrac{db}{b} = -P_0V_0\left[\ln\left(V_0^2 - a^2x^2\right)\right]$

$P_0V_0\ln\left(\dfrac{V_0^2}{V_0^2 - a^2x^2}\right)$ $\qquad ax = \dfrac{al}{4} and \dfrac{al}{2} = V_0$

$ax = \dfrac{V_0}{2} \therefore W = P_0V_0\ln\left(\dfrac{4}{3}\right)$

Since $dU = 0, d\theta = dW \Rightarrow Q = W = nRT\ln\left(\dfrac{4}{3}\right)$

If walls are insulated but piston is conducting the temperature of gas in both the chambers remains the same. Let the rise in temperature be dT.

$\therefore \Delta U = nC_V dT$

$dU = -dW$ but dW = Negative since work is done on the gas

$2nC_V dT = dU = -nRT\int\dfrac{db}{b}$

$2\dfrac{C_V}{R}\int_{T_1}^{T_2}\dfrac{dT}{T} = dU = -\int\dfrac{db}{b} \Rightarrow 2\dfrac{C_V}{R}\ln\left(\dfrac{T_2}{T_1}\right) = -\ln\dfrac{3}{4}$

$\left(\dfrac{T_2}{T_1}\right)^{\frac{2C_V}{R}} = \dfrac{4}{3}, C_V = 2R$

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$$\frac{T_2}{T_1}=\sqrt{\frac{4}{3}}\Rightarrow T_2=T_1\frac{2\sqrt{3}}{3}=1.154T_1$$

$$\Delta U=2nC_V\left(T_2-T_1\right)=2nR(62)=124nR$$

7. Liquid A apply pressure in downward direction.

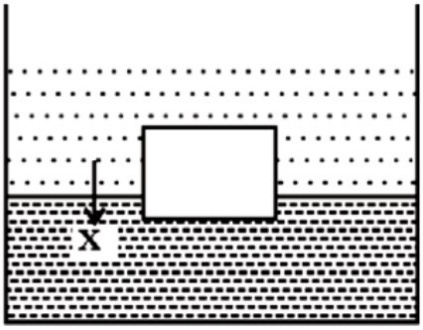
$$\rho_S Vg=mg=\rho\frac{2V}{3}g+2\rho\frac{V}{3}g\rho_S=\frac{4\rho}{3}$$

As atmospheric pressure increases then pressure at every point inside liquid increases, so force by liquid B increase.

8. As depth of block surface from top increases, so force from both liquid increases.

$$F_{net}=F_b-mg$$

$$=2\rho\left[\frac{V}{3}+xA\right]g+\rho\left[\frac{2V}{3}-xA\right]g-mg$$

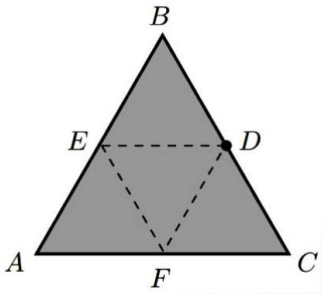


$$F_{net}=\rho Agx$$

So block perform SHM.

$$T.P=2\pi\sqrt{\frac{m}{\rho Ag}}=2\pi\sqrt{\frac{\frac{4}{3}\rho a^3}{\rho a^2 g}}=2\pi\sqrt{\frac{4}{3}\frac{a}{g}}$$

9,10.



By the principle of superposition, the potential ϕ_2 is equal to twice the potential of the vertex of an equilateral triangular plate with side a and surface charge density σ. If the dimensions of the plate are doubled, maintaining the surface charge density, then the potential of each point will also increase by a factor of 2 . To prove this fact, you can break the original plate into small parts, which can be considered point charges. As a result of scaling, the area of each part will increase by 4 times, which means the charge will increase by 4 times, and the distance to each part will increase by 2 times. Considering that the potential of a point charge is equal kq/r, we obtain that the potential created by each small part will increase by 2 times, which means that the total

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potential will double. This means that the potential of the point C is 2 times greater than the potential of the top of the plate with side $a, \phi_C = \phi_2$.

Considering that the potential of a point charge is equal kq/r, we obtain that the potential created by each small part will increase by 2 times, which means that the total potential will double. This means that the potential of the point C is 2 times greater than the potential of the top of the plate with side $a, \phi_c = \phi_2$.

Let's find the potential of the point D . To do this, mentally divide the triangle into 4 triangles with sides a. Note that AEDF - this is the original rhombus, which creates D a potential at the point $\varphi 1$. To it you need to add the potentials created by triangles BDE and DF C (Fig). The potential of each of them at the vertex is equal to $\phi_2 / 2$.

From here $\phi_D^{'} = \phi_1 + \phi_2$.

After removing the central triangle, the potential at point D decreased by ϕ_2 and became equal to $\phi_D^{'} = \phi_1 + \frac{\phi_2}{2}$.

To find the potential of the point C of the "leaky" plate, you need to find out what potential the triangle created in it DEF. Consider a rhombus CF ED . Its potential at point C is equal to ϕ_1 and is the sum of the potential created by the triangle DEF and the potential created by the triangle F DC and equal to $\phi_2 / 2$. Then the triangle DEF creates a potential at point C. This means that after removing the central triangle, the potential of point C will become equal

11. $$g = \frac{GM}{R^2} = \frac{6.6742 \times 10^{-11} \times 6 \times 10^{24}}{6.4^2 \times 10^{12}} = 9.77666$$

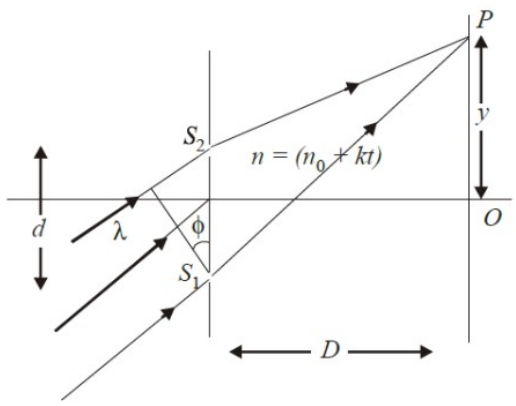
$$\frac{dg}{g} = \frac{dG}{G} + \frac{dM}{M} + \frac{2dR}{R}$$

$$= \frac{0.001}{6.6742} + 0.0001 + 2 \times 0.0002 = 0.00065$$

$$\therefore g = (9.77767 \pm 0.0007)\, m/s^2$$

12. $(A)\, S_1P - S_2P = \frac{dy}{D}$ $\qquad \Delta x = (n_0 + kt)\frac{dt}{D} - d\sin\phi = 0$

For central maxima. $\qquad \therefore y = \frac{D\sin\phi}{n_0 + kt}(y - coo$



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(B) $\frac{-kD\sin\phi}{(n_0+kt)^2}$ = velocity of central maximum

(C) For central maxima to be formed at O

$$n'\left(\frac{n}{n'}-1\right)b = d\sin\phi$$

Here $n' = n_0 + kt, n =$ refractive index of plate.

$$n = n_0 + kt + \frac{d\sin\phi}{b}$$

13. Ratio of $\cos(\omega_x t + \phi_x)$ and $\sin(\omega_y t + \phi_y)$ should be non zero finite constant.

14. Conceptual

15. Initially parabolic path means escape velocity & hence $(TE = 0)$

For perihelion $a(1-e) = R + h = \frac{3R}{2} \Rightarrow a = 3R$

Also, for elliptical path

$$T.E. = -\frac{GMm}{2a} = \frac{-GMm}{6R} \Rightarrow \text{change in } TE = \frac{GMm}{6R}$$

For aphelion: $a(1+e) = R + h = \frac{3R}{2} \Rightarrow a = R$

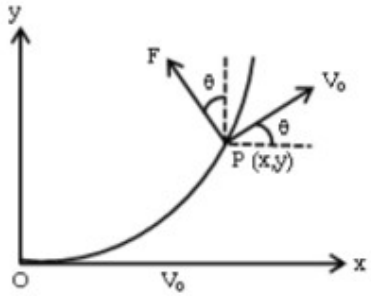
$$\&T.E. = -\frac{GMm}{2a} = -\frac{GMm}{2R} \Rightarrow \text{change in } TE = \frac{GMm}{2R}$$

16. The maximum energy of photon depends on the energy of electrons incident.

17. $\omega_1 = \sqrt{\frac{1200}{3}} = 20\omega_2 = \sqrt{\frac{1200}{27}} = \frac{20}{3}$

$$\omega_1 = 3\omega_2 t = \frac{\pi/2}{\omega_2} = \frac{3\pi}{2(20)} = \frac{6\pi}{80}$$

18. Particle moves in xy plane. Its path is not a circle because B is not uniform. However,



speed of the particle remains constant at V_0. Let velocity at point $P(x,y)$ be inclined to x axis at an angle θ. Force will be perpendicular to velocity, inclined at θ with y direction.

$$\therefore\ ma_y = F\cos\theta \Rightarrow m\frac{dV_y}{dt} = (B_0 x)qV_0\cos\theta$$

$$\Rightarrow m\frac{dV_y}{dx}\cdot\frac{dx}{dt} = B_0 q(V_x)\left[\therefore V_0\cos\theta = V_x\right]\frac{dx}{dt} = V_x\ But\ \frac{dV_y}{dx} = \frac{B_0 q}{m}x$$

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$$\Rightarrow \int_0^{v_0} dV_y = \frac{B_0 q}{m}\int_0^{x_0} x dx$$

Where $x_0 =$ maximum x co-ordinate of the particle where velocity becomes parallel to y axis.

$\therefore V_0 = \frac{B_0 q}{2m}x_0^2 \Rightarrow x_0 = \sqrt{\frac{2mv_0}{B_0 q}}$, Magnetic force at $x = x_0$ is $F = qV_0(B_0 x_0)$

$$\therefore F = qV_0 B_0\sqrt{\frac{2mV_0}{B_0 q}} = v_0\sqrt{2mqB_0 v_0}$$

If radius of curvature of path if R then $\frac{mv_0^2}{R} = F \Rightarrow \frac{mV_0^2}{R} = v_0\sqrt{2mqB_0 v_0} \Rightarrow R = \sqrt{\frac{mV_0}{2qB_0}}$

19. Initially let no. of molecules of $A = N_A$ and that of $B = N_B$

$$\frac{\lambda N_A}{\lambda N_A + 2\lambda N_B} = \frac{25}{100}$$

$$\therefore 100\lambda N_A = 25\lambda N_A + 50\lambda N_B \Rightarrow N_A = \frac{2}{3}N_B$$

Now, $N_A' = N_A e^{-\lambda t} = \frac{2}{3}N_B e^{-\lambda t}$

$$N_B' = N_B e^{-2\lambda t}$$

$$\frac{\frac{2}{3}\lambda N_B e^{-\lambda t}}{\frac{2}{3}\lambda N_B e^{-\lambda t} + 2\lambda N_B e^{-2\lambda t}} = \frac{75}{100} = \frac{3}{4}$$

$$p\frac{1}{1+3e^{-\lambda t}} = \frac{3}{4}$$

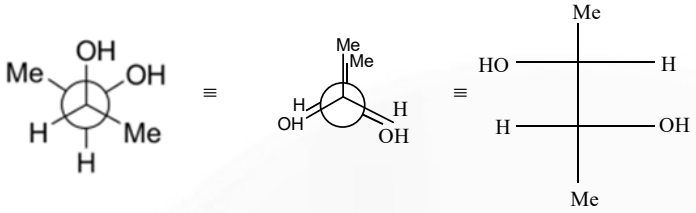
$$p_1 = 9e^{-\lambda t} p e^{\lambda t} = 9 \; So, t = \frac{\ln(9)}{\lambda} hr = \frac{2\ln(3)}{\ln(3)} hr = 2hr$$

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CHEMISTRY

20. A and B are meso
C is 2S, 3S;
D is 2R, 3R - due to intramolecular H – bond it is stable



21. For a mix of the weak acids $\left[H^+\right]=\sqrt{\sum K_i C_i + K_w}$

$\therefore \left[H^+\right]=\sqrt{10^{-8}+10^{-6}+2\times10^{-8}+10^{-10}+10^{-7}+10^{-4}}$

i.e., $\therefore \left[H^+\right]>10^{-3}$ & $pH<3$

For $0.1M\ NaB$ $pH=7+\frac{1}{2}\left(pK_a+\log C\right)=9.5$

$0.1M\ HE$ *and* $0.01HB$ have same pH values.

22. $r_n \alpha \frac{n^2}{Z}$ $\quad KE_n=\left|E_n\right|\alpha\frac{Z^2}{n^2}$

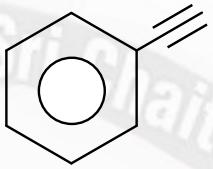
$V_n \alpha \frac{Z}{n}$ $\quad \left|\vec{L}\right|\alpha n$

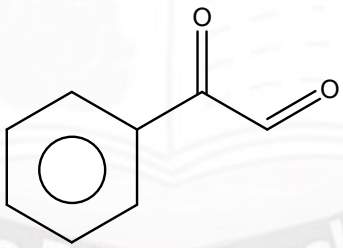
23. $\Rightarrow\left[Cr\left(H_2O\right)_4\left(en\right)\right]^{3+} \xrightarrow[-2H_2O]{+2NH_3} \left[Cr\left(H_2O\right)_2\left(NH_3\right)_2\left(en\right)\right]^{3+}$

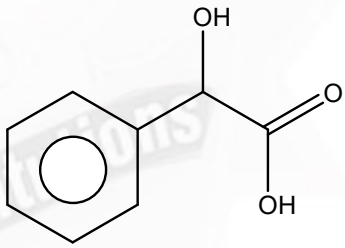
μ_{eff} =3.87 B.M
G.I=0
O.I=0

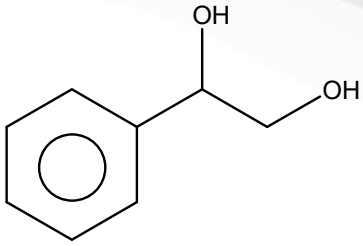
μ_{eff} =3.87 B.M
G.I=3
O.I=4

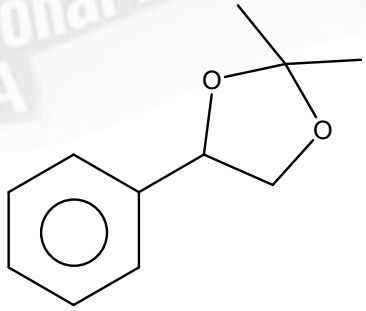
24, 25.

P is 

Q is 

R is 

S is 

T is 

X = 2, Y = 5

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26, 27.

$FeSO_4(aq)+Cr_2(SO_4)_3(aq)$

$\downarrow (NH_4)_2S$

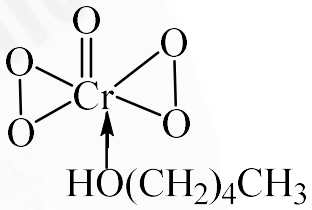
Precipitates: $FeS\downarrow + Cr(OH)_3\downarrow$ → Excess Na_2O_2 Solution

- Precipitate: $Fe(OH)_3\downarrow + Na_2SO_4(aq)$
- Filtrate: $Na_2CrO_4(aq)$

Filtrate: $SO_4^{2-}(aq)+NH_4^+(aq)$ → $BaCl_2$

$BaSO_4\downarrow$ White → dil HCl → Not soluble

$$\Rightarrow Na_2CrO_4(aq)+H^+(aq)+H_2O_2(2\text{-}3\%) \xrightarrow{\text{Amyl Alcohol}}$$



$$\Rightarrow Fe(OH)_3\downarrow + HCl \xrightarrow{\text{warm}} FeCl_3(aq)$$

$$\Rightarrow FeCl_3(aq)+SCN^-(aq) \longrightarrow [Fe^{III}(H_2O)_5(SCN)]^{2+}$$

$\Rightarrow a=6, b=8$

$\Rightarrow$ No. of unpaired electrons in $\left[Fe(H_2O)_5(SCN)\right]^{2+}=5$

28. Equation $CH_3COOH \rightleftharpoons \underset{C(1-\alpha)}{CH_3COO^-} + \underset{C\alpha}{H^+}$

$$K_a=\frac{\alpha.\left[H^+\right]}{1-\alpha} \Rightarrow 10^{-5}=\frac{0.5\left[H^+\right]}{0.5} \Rightarrow \left[H^+\right]=10^{-5}\,M$$

29. Initial $\left[H^+\right]=\sqrt{K_a-C}=10^{-3}$

Initial $pH=3$

Final pH required 56 i.e., $\left[H^+\right]=10^{-6}$.

$$K_a=\frac{\left[CH_3COO^-\right]\left[H^+\right]}{\left[CH_3COOH\right]} \Rightarrow 10^{-5}=\frac{\alpha\times 10^{-6}}{1-\alpha} \Rightarrow \frac{\alpha}{1-\alpha}=10 \Rightarrow \alpha=\frac{10}{11}$$

$$\text{Also, } K_a=\frac{C\alpha^2}{1-\alpha} \Rightarrow 10^{-5}=\frac{C\times\left(\frac{10}{11}\right)^2}{1-\frac{10}{11}} \Rightarrow C=11\times 10^{-7}\,M$$

$M_1V_1=M_2V_2 \Rightarrow 0.1\times 11=11\times 10^{-7}V_2 \Rightarrow V_2=10^6\,L.$

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30. A) $\beta-(1,4)-$ galactosidic linkage present

B) With CH_3OH / HCl gas only hemiacetal $-OH$ will react but not ordinary alcohol

D) Cellulose has $\beta-(1,4)-$ glucosidic linkage

31. B) E_2 product is major

C) Unhindered 1°- alkyl halide with W.B, weak nucleophiles like H_2O, ROH no reaction

D) With $ClSO_3H$, Chlorosulphonation take place

32. $\frac{r_{A^+}}{r_{B^-}} > 0.732 \Rightarrow CsCl$ type structure.

33. Micelle formation starts above Kraft's temperature.

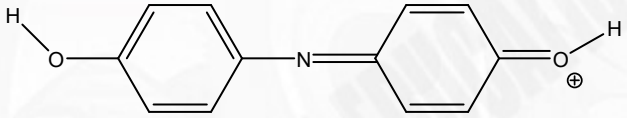
34. (i) $SO_3(g)+H_2O \rightarrow H_2SO_4 \xrightarrow{SO_3(g)} H_2S_2O_7$

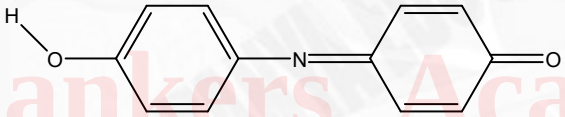
(ii) $SO_3(g)+H_2O_2 \rightarrow H_2SO_5 \xrightarrow{SO_3} H_2S_2O_8$

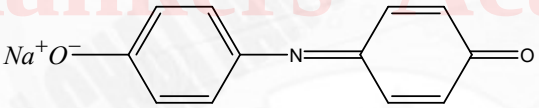
$\Rightarrow$ Sulphur atoms are in same oxidation states in all oxyacids produced.

35. Molten impure metal is exposed to atmospheric air or dioxygen and impurities are preferably oxidised.

36. It is Libermann's nitroso test for phenol

P is  D.U = 9

Q is  D.U = 9

R is Na^+O^-  D.U = 9

37. $P_s = 600 torr \quad \Rightarrow \quad x_A^0 P_A^0 + x_B^0 P_B^0 = 600$ and $\frac{y_A}{P_A^0} + \frac{y_B}{P_B^0} = \frac{1}{600}$.

These equations given $x_B = \frac{1}{3}$ and $y_B = \frac{5}{9}$

$l_A + v_A = l_B + v_B; \frac{l_A}{l_B} = 2 ____(2); \frac{v_A}{v_B} = \frac{4}{5} ____(3) \quad \Rightarrow \quad l_A = 2l_B \Rightarrow v_A = 0.8v_B$

Equation $(1) \Rightarrow \; 2l_B + 0.8v_B = l_B + v_B \Rightarrow \frac{v_B}{l_B} = 5 ____(4)$

We need $\frac{v_A + v_B}{l_A + l_B} = \frac{1.8v_B}{3l_B} = 3.$

38. $\Rightarrow Ni^{2+}(aq)+4KCN \rightleftharpoons 4K^+(aq)+\left[Ni(CN)_4\right]^{2-}$

$\Rightarrow x = 0, y = 4$

$\Rightarrow$ Electronic distribution of Ni^{2+} in $\left[Ni(CN)_4\right]^{2-} : \left[d_{yz}^2 = d_{xz}^2\right] < d_{z^2}^2 < d_{xy}^2 < d_{x^2-y^2}^o$

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MATHEMATICS

39. It is possible when $x^3+x^2+k=x-2$ has three distinct solution

$f(x)=x^3+x^2-x+k+2=0$

$f'(x)=3x^2+2x-1=3(x+1)\left(x-\frac{1}{3}\right)$

$\Rightarrow f(-1).f\left(\frac{1}{3}\right)<0$

$(k+3)\left(k+\frac{49}{27}\right)<0 \Rightarrow \left(-3,-\frac{49}{27}\right)$

41. L: Passes through $(-1,-6)$

Let slope $m\left(m=\frac{b}{a}\right)$

Line : y = mx + m – 6

For tangent $m-6=\frac{2}{m}$

$m^2-6m-2=0 \begin{matrix} / & m_1 \\ \backslash & m_2 \end{matrix}$

$2m^2+6m-1=0 \begin{matrix} / & 1/m_1 \\ \backslash & 1/m_2 \end{matrix}$

$\therefore \Sigma\frac{a}{b}=-3$

42. $\frac{x^2}{16}-\frac{y^2}{4}=1$

$\left(5,\frac{3}{2}\right)$ lies on the curve.

tangent at A

$\frac{5x}{16}-\frac{3}{2}\left(\frac{y}{4}\right)=1 \Rightarrow 5x-6y=16$

tangent at B

5x – 6y = - 6

Normal at A

6x + 5y = λ

$A\left(5,\frac{3}{2}\right)80+\frac{15}{2}=1$

$\Rightarrow \lambda=\frac{75}{2} 6x+5y=\frac{75}{2}$

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Similarly normal at B is $6x+5y=\frac{-75}{2}$

Distance between tangent is $2r=\frac{32}{\sqrt{25+36}}$

$\therefore r=\frac{16}{\sqrt{61}}$

Radius of circular of rectangle is

$R=\sqrt{25+\frac{9}{4}}=\frac{\sqrt{109}}{2}$

43, 44.

$y-f(x_1)=\frac{f(x_2)-f(x_1)}{x_2-x_1}(x-x_1)$

Passing through point $(0, 2x_1x_2)$

$\Rightarrow 2x_1x_2-f(x_1)=-x_1\left(\frac{f(x_2)-f(x_1)}{x_2-x_1}\right)$

$\Rightarrow 2(x_2-x_1)=\frac{f(x_1)}{x_1}-\frac{f(x_2)}{x_2}$

$\Rightarrow \frac{f(x)}{x}+2x=k$

$\Rightarrow f(x)=kx-2x^2$

Given $f(1)=-1$

$\Rightarrow -1=k-2 \Rightarrow k=1$

$\Rightarrow f(x)=x-2x^2$

45, 46.

$\frac{\ell_1}{a_2}=\frac{\ell_1}{a_1}=4\Rightarrow \ell_1=4a_2, \ell_2=4a$

$\frac{\frac{k}{2}(\ell_1+a_1)}{\frac{k}{2}(\ell_2+a_2)}=2$

$4a_2+a_1=2(4a_1+a_2)=8a_1+2a_2$

$2a_2=7a_1\Rightarrow \frac{a_1}{a_2}=\frac{2}{7}$

$\alpha\frac{d_1}{d_2}, \lambda=\frac{\ell_1}{\ell_2}=\frac{7}{2}$

$\alpha=\frac{\ell_1-a_1}{\ell_2-a_2}=\frac{4a_2-a_1}{4a_1-a_2}=\frac{4a_2-\frac{2a_2}{7}}{4\left(\frac{2a_2}{7}\right)-a_2}=\frac{\frac{4}{8}-\frac{2}{7}}{\frac{8}{7}-1}=26$

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47, 48.

$$g(x)=\begin{cases}x^2-x, & x\in\left[0,\sqrt{2}\right]\\ \frac{x^4}{2}-x, & x\in\left(\sqrt{2},2\right]\end{cases}$$

$$h(x)=\begin{cases}\frac{x^3}{3}-\frac{x^2}{2}, & x\in\left[0,\sqrt{2}\right]\\ \frac{x^5}{10}-\frac{x^2}{2}+\frac{4\sqrt{2}}{15}, & x\in\left(\sqrt{2},2\right]\end{cases}$$

49. $f(x)=2e^x-2\cos x+\sin x$

50. $\left(\sum_{i=1}^{n}k_i\right)\left(\sum_{i=1}^{n}\frac{1}{k_i}\right)\geq n^2\Rightarrow n^2-5n+4\leq 0 \quad\Rightarrow n=1,2,3,4$

For $n=1$ solution set is $\{1\}$

For $n=2$ no solution

For $n=3$ solution set is $\{2,3,6\}$

For $n=4$ solution set is $\{4,4,4,4\}$

51. $f(x)=2\sin\left(x+\tan^{-1}x\right)as\,x+\tan^{-1}x-x+\cot^{-1}x=\frac{\pi}{2n}$

$$\int_0^{\frac{\pi}{4}}f(x)dx<\int_0^{\frac{\pi}{4}}2\sin 2xdx=1\Rightarrow\int_0^{\frac{\pi}{4}}f(x)dx=0$$

$f(x)<2\sin 2x\Rightarrow f(x)$ attains maxima at a value greater than $\frac{\pi}{4}$.

52. A) $000\rightarrow\frac{6\times5\times4}{8\times7\times6}=\frac{5}{14}$

$OEE\rightarrow\frac{6\times2\times1\times3}{8\times7\times6}=\frac{3}{28}$

$\frac{5}{14}+\frac{2}{28}=\frac{13}{28}$

B) Variance is greatest $\Rightarrow$ No's are 1, 3, 5

$\Rightarrow\frac{2\times2\times2\times6}{8\times7\times6}=\frac{1}{7}$

C) Variance is least $\Rightarrow 1,\frac{1}{2},\frac{2}{2},\frac{3}{3},\frac{4}{4},\frac{5}{5}$

$\Rightarrow\frac{12}{8\times7\times6}=\frac{1}{28}$

53. (A) $\int_0^T f(x)dx+\int_0^e g(x)dx=e-0\Rightarrow\int_0^e g(x)dx=e-I$

(B) $\int_0^1 e^x\left(\sqrt{x}+\frac{1}{2\sqrt{x}}\right)dx=e^x\sqrt{x}\Big|_0^1\Rightarrow I+\frac{1}{2}\int_0^1\frac{f(x)}{x}dx=e\Rightarrow\int_0^1\frac{f(x)}{x}dx=2(e-I)$

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C) $\because \sqrt{x} > x, \therefore \int_{-}^{1} e^x \sqrt{x}\,dx > \int_{-}^{1} e^x dx = \left(xe^x - e^x\right)\Big|_0^1 = 1$

54. $\because \cot\theta - \frac{\sin\theta}{\cos\theta} = \frac{\cos 2\theta}{\sin\theta\cos\theta}$ _____(1)

$\frac{\cos 2\theta}{\sin\theta\cos\theta} - \frac{\sin 2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta}\left(\frac{\cos 3\theta}{\sin\theta}\right)$ _____(2)

$\frac{\cos 3\theta}{\cos^2\theta\sin\theta} - \frac{\sin 3\theta}{\cos^2\theta} = \frac{1}{\cos^3\theta}\left(\frac{\cos 4\theta}{\sin\theta}\right)$ _____(3)

$\frac{\cos n\theta}{\cos^{n-1}\theta\sin\theta} - \frac{\sin(n\theta)}{\cos^n\theta} = \frac{1}{\cos^n\theta}\left(\frac{\cos(n+1)\theta}{\sin\theta}\right)$.....(n)

Adding (1), (2), (3), ….. (n), we get

$\cot\theta - \left[\frac{\sin\theta}{\cos\theta} + \frac{\sin 2\theta}{\cos^2\theta} + \frac{\sin 3\theta}{\cos^3\theta} + ... + \frac{\sin n\theta}{\cos^n\theta}\right] = \frac{1}{\cos^n\theta}\frac{\cos(n+1)\theta}{\sin\theta}$

$\therefore\ p = 1, q = (n+1), r = 1, s = n$

55. Clearly 1st element of $A_{10} = a_{1^2+2^2+\ldots+9^2+1} = a_{286}$

And last element of $A_{10} = a_{1^2+2^2+\ldots+10^2} = 385$

$\therefore\ T_r(A_{10}) = [\log_2 286] + \ldots + [\log_2 385]$

$= 8 + 8 + \ldots + 8 = 80$

56. $a = (x+2+iy)(x-iy+2i)$

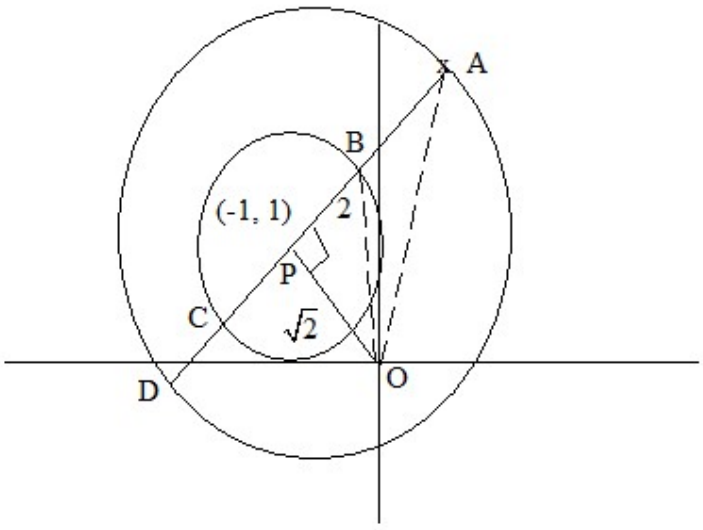
$= x^2 + 2x + y(y-2) + i(xy - xy + 2x - 2y + 4)$ is purely real $x - y + 2 = 0$

And $x^2 + y^2 + 2x - 2y \in [1, 2]$

$1 \le (x+1)^2 + (y-1)^2 \le 4$

$x = y + 2$

$(y+3)^2 + (y-1)^2 = 4$



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$|Z|_{\max} = OA = \sqrt{OP^2 + AP^2}$

$= \sqrt{2+4} = \sqrt{6}$

$|Z|_{\min} = OB = \sqrt{OP^2 + PB^2}$

$= \sqrt{2+1} = \sqrt{3}$

57. $\alpha + \beta = \tan\theta - 2$……..(1)

$\alpha\beta = -\tan\theta - 1$……..(2)

$(1)+(2) \Rightarrow \alpha + \beta + \alpha\beta = -3 \Rightarrow (\alpha+1)(\beta+1) = -2$

Hence either $\alpha+1=-2$ and $\beta+1=1$ or $\alpha+1=-1$ and $\beta+1=2$

i.e., either $\alpha=-3$ and $\beta=0$ or $\alpha=-2$ and $\beta=1$

if $\alpha=-3$ and $\beta=0 \Rightarrow \tan\theta=-1 \Rightarrow \theta = \frac{3\pi}{4}, \frac{7\pi}{4}$

if $\alpha=-2$ and $\beta=1 \Rightarrow \tan\theta=1 \Rightarrow \theta = \frac{\pi}{4}, \frac{5\pi}{4}$

Hence sum of all possible values of θ is $0\frac{\pi}{4} + \frac{3\pi}{4} + \frac{5\pi}{4} + \frac{7\pi}{4} = \frac{16\pi}{4} = 4\pi$.

Hence $k = 4$

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